

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTESWINTER SEMESTER, 2024-2025
FULL MARKS: 75**CSE 4511: Computer Networks****Programmable calculators are not allowed. Do not write anything on the question paper.**Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. Consider the network for "AllIutians," a startup based in Gazipur, which has been assigned the IPv4 address block 200.10.20.0/24. The network administrator must allocate IP addresses to the following departments based on their host requirements: Engineering (55 hosts), Sales (28 hosts), Management (10 hosts), and a WAN Link (2 IPs). **(29 Marks)**
 - a) Design an optimal addressing scheme using Variable Length Subnet Masking (VLSM) to satisfy the requirements. Explicitly state the Subnet Mask, Network Address, and Broadcast Address for all the departments. 3 × 3
(CO4)
(PO3)
 - b) Calculate the total number of assignable IP addresses in the original /24 block versus the total number of IPs actually allocated to the subnets. Quantify the "wasted" IPs and justify if this remainder is sufficient for a future department requiring 20 hosts. 5
(CO1)
(PO1)
 - c) A new client connects to the Engineering subnet and requests an IP address. Briefly explain the sequence of DHCP messages used when a client joins the network. Which of these messages are broadcast and why? 5
(CO1)
(PO1)
 - d) The startup utilizes Network Address Translation (NAT) to interface with the Internet. Suppose an internal host (192.168.1.5:4000) sends a packet to a web server (8.8.8.8:80), explain how the router modifies the header (Source IP/Port) in the outgoing packet and how it knows where to send the incoming reply. 5
(CO1)
(PO1)
 - e) If the "AllIutians" network scales to 50 routers, which protocol between OSPF and RIP would offer faster convergence and lower bandwidth overhead? Justify your choice. 5
(CO2)
(PO2)
2. Suppose IUT CSE laboratory network relies on Layer 2 Ethernet switches to interconnect various departmental workstations. **(18 Marks)**
 - a) A switch receives an Ethernet frame with Source MAC AA:AA and Destination MAC BB:BB. Assume the switch's forwarding table is currently empty. Explain the steps the switch takes to handle this frame. Explain the reason of this table having a timeout (aging time). 5 + 3
(CO1)
(PO1)
 - b) Student A wants to send data to Student B on the same LAN but only knows B's IP address. Explain how ARP resolves this. 5
(CO1)
(PO1)
 - c) You configure the switch to separate traffic into VLAN 10 (Students) and VLAN 20 (Faculty). Explain the concept of a Broadcast Domain. How does creating these two VLANs affect the number of broadcast domains? 5
(CO2)
(PO2)

3. a) An IPv4 datagram with a Total Length of 4000 bytes (including a standard 20-byte IP header) arrives at a router. The datagram must be forwarded onto a link with an MTU (Maximum Transmission Unit) of 1500 bytes. The router must fragment this packet. 9 + 5
(CO3)
(PO2)
- i. Calculate and list the values for the following fields for each of the resulting fragments: Total Length, More Fragments (MF) Flag (0 or 1), Fragment Offset
 - ii. Why is the Identification (16-bit) field critical in this process? What would happen at the destination if two different source hosts accidentally used the same Identification number for their packets simultaneously?
- b) A router receives a packet with TTL = 1. Explain exactly what the router would do now? 6
(CO1)
(PO1)
- c) Compared to IPv4, several header fields have been removed and several new fields have been introduced in IPv6. Identify the major IPv4 header fields that were dropped in IPv6 and explain why they were dropped and how their functionalities are handled in the IPv6 protocol. Additionally, discuss the newly added IPv6 header fields and justify the purpose of each with appropriate reasoning. 8
(CO1)
(PO1)

Appendix for CSE 4511

A Reference: Packet & Header Formats

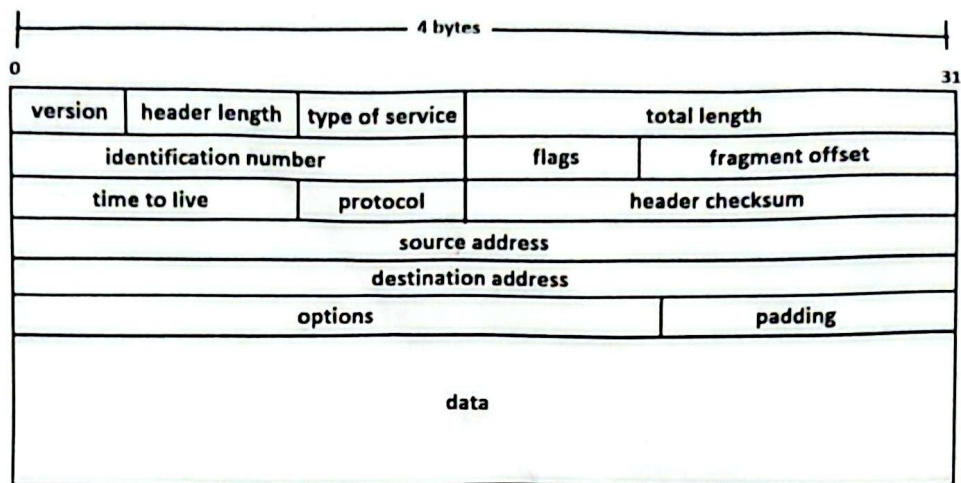


Figure 1: IPv4 Packet Format

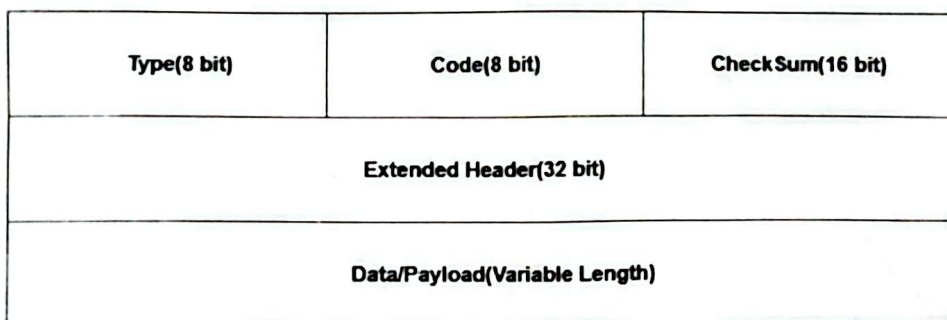


Figure 2: ICMPv4 Packet Format

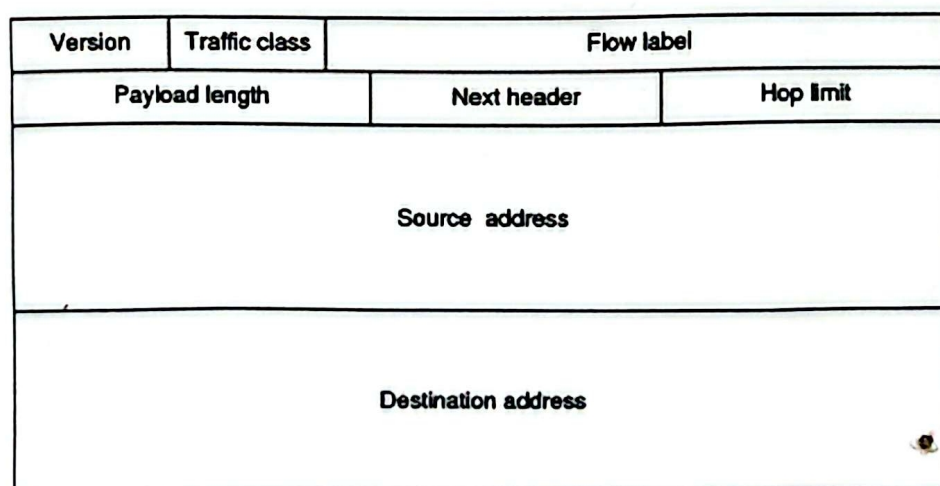


Figure 3: IPv6 Header Format